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Memory over network performance. A brief comparison of GBN and SR ARQ protocols

October 2, 2026

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Abstract: lorem

Keywords: IoT, GBN, Go Back N, Selective Repeat, Memory constraints

1 Introduction

Automatic-repeat-request protocols (ARQ) are well utitilized in data transfer to ensure data is transferred safely and in order. The three basic protocols for ARQ are stop-and-wait, Go-Back-N (GBN) and Selective repeat (SR) (Lu and Chang 1993). This paper focuses on GBN and SR, comaparing their advantages and disadvantages.

Go-Back-N and Selective repeat are both defined as "Sliding window" ARQ's by (Fairhurst and Wood 2002), meaning they both number every frame with a unique sequence number and can have multiple frames in fligth at a time, unlike in classical Stop-and-Wait protocols which only transmit one frame and wait for an acknowledgement before sending another. GBN and SR themselves differentiate in the fact that in GBN a retransmission transmits all packets from error point and in SR only erroneous packets are retransmitted (Calabro, Marchetti, and Paratore 2025).

2 Comparison

2.1 Network performance

While both GBN and SR are both classified as effective ARQ protocols when compared to older SW protocols(Weldon 1982;Shah, Ara, and Matsumoto 2005) there is still a difference between them. The main inefficiency in GBN comes from the fact that by design it needs to resend all packets up to the error point as pointed out by Shah, Ara, and Matsumoto 2005. Selective repeat is considered more efficient in this regard due to it only resending erroneous packets. This efficiency however does not come without a cost, the more complex logic is more difficult to implement and it requires more computing power from the devices communicating, this might lead to higher energy usages in IoT devices in which energy usage is a very important aspect. In addition SR protocols provided efficiency is also more theoretical and in some cases with smaller packets and low latency requirements even a regular Stop and wait protocol and achieve comparable network performance with much less overhead. (Calabro, Marchetti, and Paratore 2025).

2.2 Memory performance

Due to GBN's less complex architecture it's memory performance is much better than SR. A system using selective repeat needs to keep track of all in flight packets (Weldon 1982) which almost always requires the usage of a *Map* data structure leading to very high memory costs. In addition every packet needs to have its own timer which creates even more memory overhead. Onece again this efficiency of memory is very noticeable in devices with lower amounts of memory. A GBN system's comlexity is much lower, for example a receiver device using a GBN protocol does not need to store all received packages before delivering them since it will always deliver packets in order. In a SR system packets might be acknowledge in a different oreder meaning they can only be delivered once the entire sequence

3 Conclusions

While Selective repeat is more efficient in network performance it has some major drawbacks due to its high complexity and resource usage. The Go-back-N protocol on the other provides the efficiency of a sliding window ARQ without the high resource usages of a SR protocol. For these reasons a Selective repeat system should be used when resources are not a concern or when data packets are large and network conditions high. A GBN protocol is well suited for devices with smaller memory and energy capabilities where the potential slower network performance is not a concern.

Bibliography

Calabro, Antonello, Eda Marchetti, and Maria Teresa Paratore. 2025. “Evaluating Use of ARQ Strategies in Communication Protocols for Search and Rescue.” In *WEBIST,* 59–70.

Fairhurst, G., and L. Wood. 2002. *Advice to link designers on link Automatic Repeat reQuest (ARQ).* RFC 3366, 3366, September. https://doi.org/10.17487/RFC3366. https://www.rfc-editor.org/info/rfc3366/.

Lu, D.-L., and J.-F. Chang. 1993. “Performance of ARQ protocols in nonindependent channel errors.” *IEEE Transactions on Communications* 41 (5): 721–730. https://doi.org/10.1109/26.225487.

Shah, Alam Mohammad, Shawkat Shamim Ara, and Mitsuji Matsumoto. 2005. “An improved selective-repeat ARQ scheme for IrDA links at high bit error rate.” In *Proceedings of the 4th International Conference on Mobile and Ubiquitous Multimedia,* 37–42. MUM ’05. Christchurch, New Zealand: Association for Computing Machinery. ISBN: 0473106582. https://doi.org/10.1145/1149488.1149495. https://doi-org.ezproxy.jyu.fi/10.1145/1149488.1149495.

Weldon, E.J. 1982. “An Improved Selective-Repeat ARQ Strategy.” Cited by: 133, *IEEE Transactions on Communications* 30 (3): 480–486. https://doi.org/10.1109/TCOM.1982.1095497. https://www.scopus.com/inward/record.uri?eid=2-s2.0-0020102018&doi=10.1109%2fTCOM.1982.1095497&partnerID=40&md5=a62285287a601a28660249f391f401a6.